Compositional Perfectionism: Moral Framework for Experiential Landscapes

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Table of Contents

1. I. The Problem with Welfare
2. II. The Moral Object
3. III. Landscape Navigability
4. IV. Collapsed-Attractor Suffering
5. V. The Lower Bound
6. VI. Compositional Beauty
7. VII. The Lexicographic Structure
8. VIII. The Retribution Fork
9. IX. Local Landscapes and Global Surfaces
10. X. Non-Fungible Cultural Value
11. XI. Structural Impoverishment Without Suffering
12. XII. Voluntary Suffering and the Navigability Criterion
13. XIII. Poverty as Landscape Collapse
14. XIV. Animal Consciousness and Industrial Signal Collapse
15. XV. Death, Creation, and Signal Duration
16. XVI. Substrate Independence and Artificial Consciousness
17. XVII. The Proliferation Problem
18. XVIII. Relation to Existing Moral Traditions
19. XIX. Objections and Responses
20. XX. Formal Summary

1 I. The Problem with Welfare

We begin with a deliberately simple contrast. Suppose two lives contain the same balance of pleasure over pain, the same average hedonic tone, perhaps even the same reported satisfaction. On the standard welfarist picture, there is little left to ask. The moral quantity has been measured; the ledger balances.

But from the inside, this is plainly insufficient. A life is not encountered as an average. It is lived as an ordered passage through modes of attention, expectation, frustration, discovery, memory, effort, and change. Its value is not exhausted by whether its moments feel pleasant. We also care whether those moments open onto one another, whether they differentiate, whether they form a path rather than a loop.

This is the first pressure point in welfare ethics. Hedonic valence is real, and we should not pretend otherwise. Pain matters; relief matters; joy matters. Yet valence is only one coordinate in the subject's state. To identify the whole of moral value with that coordinate is to flatten the object we meant to evaluate. What disappears in the flattening is precisely the texture of a life: its dimensionality, its rhythm, its capacity for transformation.

To see the loss, hold fixed the quantity that welfarism is best at measuring. Imagine one life whose days are calm, agreeable, and nearly interchangeable: familiar meals, familiar entertainments, familiar thoughts, each returning with little variation. Now imagine another life with the same mean balance of pleasure and pain, but with greater internal articulation: apprenticeship, risk, grief, repair, intellectual surprise, aesthetic absorption, friendship under strain, projects that fail and then reconfigure the person who undertook them. If we ask only for the net hedonic sum, these two histories can be made equivalent.

But their equivalence is an artifact of the measuring instrument. The "comfortable rut" is easy to compress: once we know the pattern, we know almost everything. The "lived life" is not. It contains changes of register, transitions between modes of attention, local descents and recoveries, remembered contrasts, and acquired capacities. Its bad moments are not merely negative entries; they may alter the topology of the later path. Its good moments are not merely pleasant entries; they may disclose new ways of moving through experience.

So the utilitarian ledger records a sameness where phenomenology reveals a divergence.

This is the problem in its most elementary form. If we optimize directly for pleasant feeling, we select on a projection of experience rather than on experience itself. We reward smoothness even when smoothness is stagnation; we penalize disturbance even when disturbance is the condition of growth. A theory that sees only hedonic altitude cannot distinguish a plateau from a journey, or anesthesia from peace, or repetition from mastery.

Our claim, then, is not that pleasure is morally irrelevant. It is that pleasure is one coordinate in a larger subject-side dynamics. A conscious life has shape. It has dimensionality: the number of distinct ways the subject can attend, care, infer, remember, and revise. It has a compression profile: whether a long interval contains irreducible experiential structure or can be summarized as "more of the same." It has navigability: whether the person can find gradients, exits, challenges, and transformations within the landscape they inhabit.

Once this is in view, welfare ethics must change its object. The moral question is not simply how good the moments feel, but what sort of trajectory the moments compose. Comfort can be part of such a trajectory; indeed, without intervals of rest many paths break down. But comfort made sovereign becomes a trap. The aim is not maximal pleasantness. The aim is a life, and eventually a world, whose experiential paths are rich enough to be worth traversing.

2 II. The Moral Object

We begin with a shift in the unit of evaluation. An episode in the world is never morally encountered as bare input. It is filtered through a subject: through capacities, memories, expectations, sensitivities, skills, fears, and forms of attention. The same external sequence can therefore generate very different inner histories. A storm, a classroom, a hospital room, a mountain path, or a line of code is not one experience considered from different angles; it becomes many experiences because it is taken up by differently configured subjects.

We may write this schematically as

$$\tau(\theta, X_{1:T}) = (S_0, S_1, \ldots, S_T),$$

where $X_{1:T}$ is the external episode, θ is the subject profile, and each S_t is the subject-side state at time t. The moral object is this ordered trajectory. It is not merely that a subject has states; it is that those states inherit from one another, accumulate structure, revise expectations, open or close possibilities, and form a path.

This is why the framework treats experience dynamically rather than atomically. A life, or any bounded episode within a life, has shape. It can widen or narrow, repeat or develop, stagnate or transform. The morally relevant object is that temporally extended form.

We should therefore resist the familiar reductions. The relevant question is not simply how pleasant S_t is, nor whether it satisfies some prior preference attributed to the subject. Hedonic tone matters, but it is only one coordinate in a richer state. Preference satisfaction also matters, but preferences are themselves formed, revised, narrowed, and expanded by the trajectory through which the subject moves.

At each moment, the subject-side state contains a structured bundle: what is sensed, what is selected for attention, what is anticipated, what violates anticipation, what is learned from that violation, what is retained in memory, and what is taken to matter. These elements are not morally interchangeable. A calm routine, an anxious discovery, a demanding apprenticeship, and a period of grief may share the same average valence while differing profoundly in experiential form.

So the object of evaluation is not a scalar extracted from the path. It is the path itself: the evolving organization of attention, expectation, surprise, memory, and value across time. To evaluate an experience is to ask what kind of movement through subject-side state space occurred, what possibilities it disclosed, what capacities it exercised, what patterns it sedimented, and what further regions of experience it made reachable or unreachable.

This formulation is deliberately neutral about the material in which the trajectory is realized. Nothing in $\tau(\theta, X_{1:T})$ requires carbon chemistry, mammalian affect, or the particular sensory channels familiar from human life. What matters is not the substrate as such, but the organized production of a subject-dependent history: a temporally coherent succession of states in which attention is allocated, expectation is formed, error is registered, memory is updated, and value is assigned.

We should be cautious here. Substrate-independence is not the claim that every information-processing system has moral standing. A thermostat has state transitions; a database has memory; a language model may have outputs that look expressive. These facts alone are insufficient. The relevant threshold is the emergence of a unified trajectory: states must belong to a persisting point of view, carry forward the consequences of prior states, and alter the field of future possibilities for that same subject.

If such a trajectory exists, however, its moral relevance does not wait for biological resemblance. Silicon, synthetic tissue, distributed computation, or unfamiliar hybrid media would all fall within the scope of the framework if they generate coherent subject-side paths. We therefore attach concern to the structure of experienced movement, not to the historical accident of the material that bears it.

3 III. Landscape Navigability

At the level of a single conscious being, we should not ask first whether the present affective reading is pleasant. We should ask whether the being inhabits a world in which movement remains possible. A life can feel mildly agreeable while becoming almost pathless: the same cues, the same responses, the same rewards, the same narrowing of attention. Conversely, a difficult episode can be morally acceptable, even valuable, when the subject can still learn, revise, attempt, recover, and discover new directions.

This is why we model welfare locally as a navigability condition on the subject's effective loss landscape. The relevant landscape is not merely the external environment. It is the environment as filtered through temperament, memory, skill, social position, bodily condition, and available concepts. A door that exists physically but cannot be perceived, afforded, or acted upon is not yet an exit in the moral sense.

The criterion is therefore geometric rather than hedonic. We are looking for room to search, signals that make search intelligible, and enough independent axes of action or attention that the trajectory need not collapse into a single repetitive orbit. In particular, the subject must sometimes be able to approach L^*, the zone in which difficulty and capacity are matched closely enough for growth rather than futility.

More precisely, we require four features. First, the landscape must have structure: the subject's attempts must encounter regularities rather than sheer noise. If nothing can be learned, then effort cannot accumulate into orientation. Second, it must contain gradients: some differences between action, attention, interpretation, or practice must make things better or worse in a way the subject can register. A world may be difficult, but difficulty becomes morally corrosive when every direction feels equally futile.

Third, the landscape must have dimensionality. There must be more than one axis along which the subject can move: bodily, social, intellectual, aesthetic, practical, spiritual, or relational. Low-dimensional lives are not necessarily painful, but they are vulnerable to collapse because the loss of one pathway can become the loss of all pathways. Finally, there must be intermittent access to L^*, the region of optimal challenge. This does not mean constant flow or permanent achievement. It means that the being is not structurally excluded from episodes in which capacity and difficulty meet.

We can summarize the condition by saying that, across sufficiently large windows, $Q(W)$ must remain above a minimum threshold: enough learnable pattern, usable gradient, effective dimension, and reachable challenge for the trajectory to remain a path rather than a loop.

Notice the consequence. A life may clear every ordinary hedonic test while still failing as a life for a conscious subject. The person may report contentment, avoid acute distress, and maintain a stable positive affective baseline; nevertheless, the trajectory may have become thin. The same satisfactions recur, the same interpretations are rehearsed, the same capacities remain unused, and the same small orbit is traversed with increasing fluency. Nothing in the affective reading alone tells us whether d_{eff} has collapsed, whether gradients have vanished, or whether the apparent peace is really a loss of reachable alternatives.

We should therefore distinguish comfort from navigability. Comfort is a local tone of experience. Navigability is a structural property of the life-space through which experience unfolds. A comfortable life can be morally deficient when it becomes too compressible, when its future is already contained in its present, when the subject no longer encounters invitations to reorganize attention or action. Conversely, an uncomfortable life can remain morally intact when it preserves movement, intelligibility, and possible transformation.

This is precisely where scalar welfare theories lose resolution. If the affective sum is acceptable, they have little left to say. Our framework says that something important may already have gone wrong. The problem is not pain, but experiential impoverishment.

4 IV. Collapsed-Attractor Suffering

We should distinguish ordinary pain or frustration from the special regime at issue here. A conscious trajectory can pass through negative regions without being morally catastrophic, provided it still has shape: alternatives remain visible, attention can reorganize, memory can integrate the episode, and the next state is not mechanically forced by the last. Collapsed-attractor suffering names the contrary case.

Formally, the pattern is conjunctive. First, valence remains negative across a sufficiently long window: $V_t < 0$ not as a momentary dip, but as the background tone of the episode. Second, effective dimension approaches its minimum: d_{eff} is low because the subject no longer explores multiple independent modes of attention, action, or interpretation. Third, exit gradients disappear. The system cannot find a direction in which effort predictably improves the local landscape. Fourth, the transition skeleton rigidifies: the same small set of states recur, with little plasticity and little capacity for redescription.

The moral importance lies in the conjunction. Negative valence alone is not enough; low dimensionality alone is not enough. The distinctive wrong occurs when aversiveness, compression, and navigational closure reinforce one another, producing a state path that no longer functions as a path in any meaningful sense.

We can see the signature most clearly in familiar human cases. In chronic pain, attention is pulled again and again toward the same bodily alarm, while projects, social meanings, and future-directed possibilities lose their organizing force. In severe depression, the world may remain objectively varied, but it is received through a narrowed interpretive channel: each event returns the subject to the same conclusion, the same incapacity, the same affective floor. In grinding poverty, practical agency can be consumed by recurrence: rent, hunger, debt, exposure, threat. The next move is dictated before it is chosen. In solitary confinement, the environment itself is engineered to remove perturbation, relation, and redescription; the mind is left to orbit its own deprivation.

These cases are not unified by intensity alone. A brief agony may be terrible without becoming this. What unifies them is the collapse of usable state-space. The subject does not merely dislike where they are; they cannot find a meaningful direction out. The local future ceases to branch. Memory stops enriching the trajectory and instead confirms the trap. Prediction error no longer opens learning, because nothing learned alters the structure of what can happen next. The experiential record becomes maximally poor: aversive, repetitive, and almost perfectly compressible.

Thus our diagnostic rule must be read structurally rather than sentimentally. We are not asking whether an episode contains pain, fear, boredom, or grief. We are asking whether the subject still inhabits a landscape: whether distinctions remain available, whether effort can discover gradients, whether attention can shift into new modes, whether the future retains more than one live contour. If these conditions remain, then even severe suffering may belong to a trajectory that can be integrated, learned from, resisted, or transformed.

Collapsed-attractor suffering begins where those conditions fail together. Negative valence supplies the aversive tone; low d_{eff} removes experiential breadth; absent gradients remove practical hope; rigid recurrence prevents the state path from becoming anything else. The framework therefore "fires" only when these features lock into a single regime. The being is not merely undergoing something bad. The being is trapped in a local experiential machine that reproduces badness while stripping away the dimensions in which response would be possible.

This is why the category has special moral force. To leave a being there is not simply to permit disvalue. It is to permit the annihilation of navigability itself.

5 V. The Lower Bound

We begin, then, with a constraint rather than an ideal. Before we ask whether the total pattern of lives is intricate, diverse, or beautiful, we ask whether any conscious trajectory has been driven into a collapsed attractor. This priority is not additive. It is not one term in a welfare function that may be outweighed by sufficiently many later gains. It is lexicographic: if the condition is violated, the optimization problem has not yet begun.

The reason is structural. A collapsed attractor is not merely an unpleasant interval within an otherwise traversable life. It is the failure of traversability itself: negative valence coupled to vanishing effective dimension, absent exit gradients, and a transition skeleton that returns the subject to the same point. In such a regime, the being is no longer moving through an experiential landscape in any morally meaningful sense; the landscape has narrowed into confinement.

We should therefore understand the lower bound as a firewall against aesthetic aggregation. It prevents us from treating one subject's immobilization as the price of another region's richness. However large the surrounding composition becomes, its evaluation is suspended until the trapped trajectory is made navigable again.

Operationally, this means that our first question is remedial rather than maximizing. We do not ask how much population-level diversity could be purchased by leaving some being where it is; we ask what would be required to reopen its landscape. The relevant intervention may be medical, social, architectural, legal, economic, or relational. Its common form is not the production of pleasure as such, but the restoration of possible motion: more than one attentional mode, credible exits, learnable structure, and intervals in which challenge can again be met rather than merely endured.

Only after this condition is satisfied do the higher-order evaluative questions become well-posed. A society may then ask how to cultivate many forms of craft, inquiry, intimacy, ritual, play, and contemplation. It may ask how to increase $d_{\mathrm{eff}}^{\mathrm{pop}}$, how to preserve rare cultural trajectories, or how to design institutions that support optimal challenge. But these are second-order questions. They presuppose that no trajectory has been abandoned in a region where experience has become repetitive injury.

Thus the lower bound is not hostile to beauty. It is the condition under which beauty remains moral rather than predatory. To compose a world while ignoring its dead zones is not composition; it is decoration over confinement.

We can put the point in its most compact form: once we permit an individual loss landscape to be collapsed for the sake of a more impressive aggregate pattern, the subject has ceased to be a center of experience in our theory and has become material for composition. That is precisely the conversion the lower bound forbids. It does not say that beauty is unimportant, nor that social life should be organized around mere safety. It says that beauty may not be purchased by converting a conscious trajectory into a fixed sacrifice.

This matters because the language of richness can otherwise become dangerously seductive. One can always describe the larger whole as more complex, more dramatic, more culturally fecund, or more historically significant. But if that description depends on someone's being held in a state with no credible exit, then the relevant complexity is parasitic on immobilization. We have not discovered a higher aesthetic order; we have hidden a dead zone inside it.

So the lower bound gives compositional perfectionism its moral spine. We may seek worlds of many registers, many forms of attention, many difficult excellences. But we seek them under a prohibition: no trajectory may be intentionally or negligently reduced to nonnavigable suffering as the medium through which the larger pattern is made beautiful.

6 VI. Compositional Beauty

Once the lower bound is secured, our question changes. We are no longer asking, first of all, how to rescue a trapped trajectory from collapse. We are asking what kind of world should exist when conscious beings are not being crushed into dead experiential zones. The answer cannot be "maximize pleasure," because pleasure is only one coordinate of the subject-side path. Nor can it be "maximize preference satisfaction," since preferences may themselves be adapted to narrow, repetitive, or impoverished conditions. Conditional on navigability, we should optimize for the beauty of the total experiential landscape.

By beauty we do not mean decoration, luxury, or aesthetic pleasure in the ordinary sense. We mean the structural richness of the whole composition of conscious lives: how many kinds of attention are realized, how many trajectories resist reduction to one another, how much of the possible space of experience is actually traversed, and how often beings encounter challenges that stretch rather than break them. A morally excellent world is therefore not merely a world with high average welfare. It is a world whose population of lives forms a rich, high-dimensional, difficult-to-compress composition. Its first measures concern the range and differentiation of the paths themselves.

The first such measure is population-level dimensionality, $d_{\text{eff}}^{\text{pop}}$. We are not counting persons, pleasures, or episodes. We are asking how many independent directions are actually being taken by the ensemble of subjective trajectories. A population in which many lives differ only by superficial content—different jobs, different entertainments, different opinions, but the same attentional posture and the same temporal rhythm—has lower dimensionality than it appears to have. By contrast, a smaller population may occupy a much larger experiential volume if its members cultivate distinct crafts, relations, perceptual disciplines, forms of memory, modes of inquiry, and styles of evaluation. The relevant question is whether adding these lives expands the span of the composition or merely adds redundant points within an already saturated subspace.

A second measure is diversity, understood not as demographic variety by itself but as distance between trajectory shapes. Two beings may have similar hedonic averages while their lives curve through state space in radically different ways: contemplation, caregiving, wilderness travel, mathematical research, ritual, play, mourning, apprenticeship. Diversity is high when such forms coexist without being forced into a single template of attention, success, or desire.

A third measure is aggregate incompressibility. Here we ask how much of the population's trajectory structure would be lost if we tried to summarize it by a small set of scripts. A world is poor, compositionally, when its lives can be described by a few repeating templates: work, consume, recover, scroll, repeat. It may contain enormous quantities of sensation and even considerable pleasure, yet remain structurally thin. Incompressibility, in the relevant sense, is not mere randomness. Noise is not a life well formed. What matters is organized irreducibility: patterns rich enough to be intelligible, but particular enough that they cannot be collapsed into one another without moral loss.

A fourth measure is the prevalence of optimal challenge, the fraction of being-moments spent near L^*. By L^* we mean the region in which a subject's capacities are genuinely recruited: not idle comfort, not crushing impossibility, but difficulty with a path through it. This measure prevents compositional beauty from becoming an aristocratic ideal in which a few extraordinary trajectories decorate a stagnant population. A beautiful world is not one where only rare geniuses, saints, explorers, or artists encounter depth. It is one in which ordinary beings, across many forms of life, repeatedly meet tasks, relations, and environments that call them into fuller motion.

7 VII. The Lexicographic Structure

We can now state the structure of the view. The evaluative order is lexicographic:

$$P_1 \succ_{\text{lex}} P_2 \succ_{\text{lex}} P_3.$$

Here P_1 is the prohibition on collapsed-attractor suffering. No conscious being may be left in a regime of sustained negative valence, dimensional collapse, absent exit gradients, and repetitive temporal enclosure. This is the lower bound on moral admissibility.

P_2 is the requirement of landscape navigability. Once collapse has been ruled out, each being must have access to a loss landscape with learnable structure, real gradients, multiple independent directions of movement, and periodic access to optimal challenge. A life need not be easy; indeed, ease may be part of the problem. What matters is that the subject can still move, learn, differentiate, and reconfigure.

P_3 is compositional beauty: the richness of the total experiential landscape. Only after the first two constraints are met do we ask how diverse, high-dimensional, incompressible, and flow-saturated the population of trajectories can become.

The order matters. We do not purchase beauty by tolerating trapped torment, and we do not purchase aggregate richness by stripping individuals of navigable terrain. The higher priority sets the admissible space within which the next priority operates.

Thus the decision procedure is sequential. We do not begin with a social objective function and assign large coefficients to the most urgent terms. We ask, first, whether the P_1 constraint is satisfied. If it is not, optimization has not yet begun; the task is remediation. We ask, second, whether each subject has a navigable field of possible development. If not, the task is repair of access, gradient, and dimensionality. Only within the region left open by these constraints do we then compare worlds by their compositional richness.

This distinction is essential. A weighted sum always permits trade: sufficiently much value in one term can compensate for loss in another. Lexicographic order denies that permission. The higher priority is not expensive; it is inviolable. It functions as a boundary condition on admissible world-making, not as a preference to be balanced against other preferences.

The metaphor of a firewall is therefore apt. It blocks a familiar moral temptation: to redescribe a being's ruin as the cost of some larger pattern. The framework allows ambition about beauty, diversity, and depth, but only after the protected conditions of conscious trajectory have been secured.

It is important, however, not to mistake lexicographic order for practical separation. The priorities are ordered as constraints, but the interventions that satisfy them often co-operate. Restoring someone from collapse does not merely remove a forbidden state; it also reopens gradients, returns dimensionality, and makes possible future contribution to the larger composition. A rescued trajectory is not only less bad. It becomes again a site of movement.

Consider stable housing, pain relief, psychiatric care, addiction treatment, or protection from violence. These are P_1 interventions when they extract a subject from a collapsed attractor. But the same measures also serve P_2, because they restore navigability: sleep, attention, trust, memory, planning, and access to effortful projects. And once the person can move again, their trajectory may add irreducible structure to P_3, since a life no longer compressed by desperation can differentiate into work, relation, play, craft, and thought.

The same is true at institutional scale. Good schools, humane cities, public health systems, libraries, studios, laboratories, parks, and rituals of civic belonging are not merely welfare devices. They are gradient-making structures. They prevent collapse, sustain navigability, and multiply the forms conscious life can take.

Thus the lexicographic structure is strict in principle but generative in application. The priorities do not compete by default. Properly designed worlds satisfy the lower constraints in ways that enrich the higher landscape.

8 VIII. The Retribution Fork

We now encounter the point at which the framework must refuse an easy simplification. Retributive justice is not merely an application case; it tests the moral grammar of the lower bound itself. If a being has intentionally destroyed the navigability of others' lives, do our protections still attach without qualification? Or may the community restrict that being's landscape in a way that would otherwise be impermissible?

Our answer is deliberately bifurcated. The framework can show what is at stake, and it can rule out certain options, but it cannot honestly pretend that all reasonable moral premises converge here. Some readers will treat the lower bound as an unconditional fact about conscious trajectories: no subject-side path may be converted into a dead zone, no matter what the agent has done. Others will understand the lower bound as part of a reciprocal civic structure: those who systematically violate the conditions of mutual navigability may lose some claims against restriction.

Both readings are coherent within the larger architecture. The disagreement concerns the source of the protection, not the diagnostic vocabulary. In either case, the crucial distinction remains between containment, deprivation, accountability, and the deliberate production of experiential collapse. That distinction is the common ground from which the fork begins.

On the first reading, the lower bound attaches to consciousness as such. It is not earned by innocence, citizenship, reciprocity, or contribution to the common good. Once there is a subject whose trajectory can collapse into the signature state, the prohibition is activated. The community may restrain dangerous agency; it may remove opportunities for further harm; it may impose routines of accountability, repair, surveillance, and exclusion. What it may not do is design the interior of a life as a pit. Punishment cannot permissibly aim at producing a low-dimensional negative attractor: monotony plus terror, pain plus hopelessness, isolation plus no exit gradient.

The intuition here is not sentimental leniency. It is structural hygiene. A deliberately maintained hell is not a morally useful counterweight to prior wrongdoing; it is an additional collapsed region in the total experiential landscape. If we justify it by appealing to satisfaction, closure, or expressive denunciation, we must ask what kind of states those satisfactions cultivate in the witnesses. Often they are themselves narrow states: fixation, cruelty rehearsal, identity organized around enemy destruction. Thus the absolute reading treats humane containment not as mercy after justice, but as a constraint internal to justice.

On the second reading, by contrast, the lower bound is not simply a property of sentience considered in abstraction. It is a norm sustained by a community of mutually vulnerable beings. To enjoy its full protection is to stand within a scheme whose central rule is: do not intentionally collapse the landscapes of others. A person who repeatedly, knowingly, and severely violates that rule may forfeit some claims against restriction. We may justifiably narrow his options, remove freedoms, impose confinement, interrupt projects, and deny forms of association that would permit renewed predation.

But forfeiture is not metaphysical erasure. The offender does not become raw material, an object, or a permissible site for aestheticized cruelty. What is forfeited is full access to the ordinary range of navigable opportunities, not the minimal standing of a conscious trajectory. The community may build walls; it may not build hells. It may reduce reach, power, privacy, and spontaneity; it may not deliberately engineer the signature of collapse: sustained negative valence joined to dimensional starvation, hopeless repetition, and absent exit gradients.

Thus the second reading is more punitive, but not unbounded. It permits deprivation as a response to destroyed trust. It forbids destruction as a response to destruction. And here the two readings meet: whatever we think about desert, no framework committed to experiential landscapes can license the manufacture of collapsed-attractor torment for any being.

9 IX. Local Landscapes and Global Surfaces

Effective altruism is most persuasive when it asks us to notice an evasion. We would rather keep moral attention local, familiar, and aesthetically pleasing; meanwhile, preventable disease, hunger, and confinement persist elsewhere. The challenge is real, and we should not soften it. Once a collapsed or nearly collapsed trajectory enters our knowledge, the claim it makes upon us is not optional.

But the further move is different. From the reality of urgent claims, effective altruism often infers a geometry of total commensurability: all goods are points on one allocative plane, all actions are bids against one another, and moral seriousness consists in choosing the globally maximal bid. On that picture, the workshop, the festival, the parish choir, the slow apprenticeship in a craft, and the shaping of stone must all justify themselves by the same metric that justifies malaria prevention.

We deny that this is the right geometry. The error is not caring too much about distant strangers; the error is flattening the moral space in which caring occurs. Human lives are not mere spending channels through which impersonal value is routed. They are situated trajectories, sustained by places, skills, relationships, rituals, and long projects. To demand that every such trajectory be continuously reorganized around a global calculator is already to damage the landscape one is trying to improve.

The distinction is easiest to see if we treat a moral demand not as a price tag but as a role in an ordered practical grammar. The claim of the feverish child is a constraint: if we are in a position to help restore navigability where a life is collapsing, we must not decorate around that fact. It blocks certain plans until it is answered. The cathedral, by contrast, is not an alternative method for maximizing the same quantity. It is a mode of enrichment: a long, locally embedded project through which attention, skill, memory, and shared form acquire depth.

Thus the question is not whether an arch is "worth more" than a mosquito net. That comparison has already accepted the wrong surface. Priority 1 governs admissibility; Priority 3 concerns the shape of flourishing within the admissible region. Once the urgent constraint has been met, or once one's appropriate share in meeting it has been taken up, the remaining question is not how to convert every gesture into relief. It is how to inhabit and elaborate a navigable form of life without denying the claims that interrupt it. A morality with only interruption cannot build; a morality without interruption becomes ornament over a pit.

Call the first posture *responsive obligation*. It is activated by a determinate encounter with collapse: knowledge, proximity, institutional participation, causal contribution, or ordinary membership in a scheme that can repair the damage. Responsiveness is not sentimental localism. It does not say that only nearby suffering matters. It says that moral emergency enters practical life as a constraint to be answered, not as a permanent command to dissolve all projects into a single maximization exercise.

Call the second posture *calculative obligation*. Here the agent must continuously ask whether each hour, euro, friendship, apprenticeship, or celebration is the highest-yield intervention available anywhere. This posture may appear more rigorous, but its experiential effect is homogenizing. It trains every subject into the same attentional stance: comparison, ranking, opportunity-cost surveillance, guilt, and redirection. If universalized, it does not merely move resources; it narrows the repertoire of human attention.

The stonemason, then, is not excused from malaria. If the crisis has entered his moral field, he must contribute to restoring navigability in the proportion properly his. But after that response, he need not become an abstract allocation device. He may return to the stone. In doing so he preserves a trajectory with its own gradients, resistances, memories, and skills. The world is better not only because fewer children die of fever, but because, in the admissible region secured by that response, there remain stonemasons.

10 X. Non-Fungible Cultural Value

A culture is not merely a container in which welfare is produced. It is a patterned way of making the world intelligible: a repertoire of attention, memory, gesture, obligation, humor, craft, mourning, celebration, and expectation. When such a form persists across generations, it stabilizes a distinctive family of subjective trajectories. It teaches its members what to notice, how to interpret surprise, where to locate value, and which transformations of the self count as growth.

This is why population size cannot be the sole measure of cultural importance. A small community may occupy a region of experiential state space that no large civilization occupies. Its contribution to the total composition is then not proportional to headcount. The relevant question is not simply how many people instantiate the form, but whether the form opens modes of consciousness, relation, and practical intelligence that would otherwise be absent from the human landscape.

The analogy with biodiversity is instructive. We do not value an ecosystem only by summing animal pleasure or counting organisms. We also care about the plurality of forms of life, because each form discloses a different solution to existence. Cultural plurality is biodiversity at the level of consciousness. It preserves distinct ways for experience to become structured, meaningful, and transmissible.

If a cultural form anchors a trajectory class C, then its disappearance is not merely a decline in welfare within C. It is the removal of C as an available form of life. Once the language is no longer spoken in use, the ritual no longer inhabited, the craft no longer taught through bodies and corrections, we do not retain the same object in a museum or archive. We retain evidence of it. The living generative mechanism has ceased.

This matters because experiential forms are not perfectly reconstructible from descriptions. A grammar of attention is carried by practice, imitation, correction, shared expectation, and the slow calibration of memory. When that system breaks, later abundance elsewhere cannot simply purchase it back. A billion flourishing lives in other forms may be very good; they do not re-open the exact region of possibility that has gone extinct. The issue is not that the lost culture contained more pleasure than alternatives. The issue is that it disclosed a distinctive way for consciousness to move.

We should therefore treat cultural extinction as a dimensional loss in the total human landscape. Compensation can increase richness elsewhere, but it cannot make the missing shape present again.

This conclusion has an important limit. Non-fungibility is not sovereignty. To say that a cultural form occupies irreplaceable experiential territory is not to say that every practice by which it maintains itself is protected. A language, ritual, craft, or homeland may have non-fungible value; it does not follow that persons may be trapped, coerced, immiserated, or rendered unnavigable in order to preserve its silhouette.

Here the lexicographic structure does decisive work. Priority 1 still forbids collapsed-attractor suffering. Cultural continuity may justify support, protection, institutional accommodation, and resistance to homogenizing pressure. It may not justify torture, permanent subordination, forced enclosure, or the deliberate destruction of individual navigability. A form of life is valuable because it opens a distinctive way for consciousness to move. If its preservation requires preventing consciousness from moving, the argument has turned against itself.

Thus both familiar reductions fail. The utilitarian is wrong to suppose that cultural loss can always be offset by enough welfare elsewhere. The unconstrained nationalist is wrong to suppose that cultural distinctiveness overrides the lower bound. We should protect non-fungible forms of life, but only under the standing condition that the beings who inhabit them retain access to navigable landscapes. Cultural value is real; it is not absolute.

11 XI. Structural Impoverishment Without Suffering

We should begin with the peculiar moral invisibility of the screen trance. It does not announce itself as agony. No one need be screaming, starving, or imprisoned. A person may report that the hours were pleasant enough, that nothing especially bad occurred, that the stream was entertaining, soothing, or useful. On ordinary welfare measures, this looks close to neutral. Perhaps even mildly positive.

But our framework asks a different question. It asks not merely how the interval felt along the valence axis, but what kind of path the subject traced through experiential space. Did the episode open new modes of attention? Did it recruit memory, skill, embodied action, social attunement, imagination, resistance, discovery? Or did it keep the subject cycling through a narrow, repeatable configuration while presenting the appearance of novelty?

This is why the phenomenon is so grave. The screen trance can impoverish experience while sparing the subject pain. It can flatten the trajectory without lowering reported happiness very much. Indeed, its comfort is part of the danger: the collapse is not felt as collapse. We are therefore looking at a form of structural damage that hedonic ethics is poorly equipped to register. The catastrophe is not mass suffering, but mass narrowing.

The diagnostic signature is therefore subtle. The external stream may be maximally varied: jokes, wars, faces, recipes, scandals, instructions, confessions, advertisements. Measured at the level of content, the entropy is high. But the subject-side operator applied to each item is nearly invariant. The eyes scan, the thumb advances, the next microstimulus arrives, and the same shallow evaluative posture is reinstalled. Surprise is local and rapidly consumed; memory is weakly integrated; action is usually reduced to selection, dismissal, or forwarding.

So the relevant contrast is not between many items and few items. It is between available information and engaged information. The screen offers enormous latent structure, but the attentional profile samples it through a narrow channel. We might write this as a large gap between environmental entropy $H(X_t)$ and trajectory dimension $d_{\text{eff}}(\tau)$. The world presented to the subject is noisy and abundant; the world actually traversed by the subject is thin.

This also explains the flat compression profile. A week of scrolling contains thousands of distinguishable objects, but the sequence of lived modes is easy to summarize: orient, register, react, advance; orient, register, react, advance. Novelty has migrated from the form of experience to the decorations passing across it.

The individual case is already morally significant, but the population-level case is different in kind. If one person spends an evening in a narrowed mode, the loss is local and perhaps reversible. If billions of persons spend the central hours of their lives in nearly the same narrowed mode, then the loss is compositional. We do not merely have many thin trajectories; we have convergence of the whole population toward a common attentional template. The relevant variable is $d_{\text{eff}}^{\text{pop}}$, and it falls even while the quantity of circulating content explodes.

This is what we mean by experiential monoculture. Human beings have always shared rituals, stories, tools, and habits, but these shared forms were embedded in different bodies, places, crafts, seasons, languages, and obligations. The screen configuration is more abstract and more portable. It installs the same posture in the subway, the bed, the classroom, the workplace, and the meal. The local landscape is replaced by a standardized interface.

A utilitarian ledger has difficulty registering the injury, because no large negative valence need appear. The reports may remain acceptable: amused, informed, distracted, relaxed. But the landscape has lost independent directions of movement. From our perspective, this is therefore not a trivial cultural complaint. It is a structural emergency: a vast reduction in the dimensionality and diversity of conscious life without the warning signal of suffering.

12 XII. Voluntary Suffering and the Navigability Criterion

We should begin by separating voluntary difficulty from damage. A life may contain pain, deprivation, humiliation, fear, or long stretches of failure, and yet be morally rich rather than morally defective. The mountaineer on the ice face, the monastic in discipline, the artist in years of frustrated apprenticeship: these are not counterexamples to the lower bound. They are cases in which negative valence is embedded inside a larger form.

What matters is not whether the local affect is pleasant, but whether the episode belongs to a traversable landscape. The climber suffers cold and danger, but the world remains articulate: holds appear, routes suggest themselves, skill matters, attention differentiates. The monastic renounces comfort, but renunciation opens a structured interior terrain of concentration, temptation, insight, and self-command. The artist endures rejection and repetition, but each failed work refines perception and changes the next attempt.

In such cases suffering is not a pit; it is resistance. It gives contour to the trajectory. We therefore need a diagnostic that can distinguish asceticism, discipline, ordeal, and meaningful risk from collapse. Voluntary suffering is morally intelligible only where the subject remains an agent moving through a landscape, not a consciousness pinned inside one.

The diagnostic is therefore structural. During the ordeal we may observe extended intervals of negative valence, perhaps even severe negative valence, but they occur within a high-dimensional state path. Attention is not reduced to one repeating point. The subject continues to encounter distinctions: better and worse tactics, deeper and shallower interpretations, more and less skillful responses. Formally, d_{eff} remains substantial; the transition skeleton is articulated rather than rigid; prediction error still updates competence and orientation.

Most importantly, there are exit gradients. The subject can stop, descend, abandon the vow, leave the studio, ask for help, or reinterpret the project. These exits need not be costless. A climber may lose the summit; a monk may lose the discipline; an artist may lose the work. But the costs are internal to a navigable landscape, not walls sealing the subject inside collapse. Voluntariness here means not a single past consent but the continuing availability of agency across time.

We should therefore ask: is the negative state embedded in a path the being can still steer, or has steering itself disappeared? In voluntary suffering properly so called, pain functions as resistance against which form emerges. The trajectory remains open, differentiated, and responsive.

This is why two intervals with the same hedonic average can have opposite moral descriptions. Suppose the reported valence curve is equally low in the climber and in the prisoner: fatigue, fear, bodily distress, humiliation before one's limits. A scalar account sees a match. We do not. In the first case the negative affect is distributed through a wide and responsive trajectory; in the second it is coupled to dimensional collapse. The signs may be alike at the level of V_t, but the state paths differ in their geometry.

The relevant contrast is therefore not pain versus pleasure, but resistance versus entrapment. Resistance preserves intelligible next moves. It solicits attention, skill, memory, and interpretation; it changes the subject by giving the subject something to do. Entrapment removes those degrees of freedom. It converts experience into recurrence without leverage, a bad state that neither teaches nor opens. Utilitarianism tends to register both cases under the same debit column, because both lower the affective sum. Compositional Perfectionism separates them by asking what kind of trajectory the suffering inhabits. The moral question is not simply, "How bad does it feel?" It is also, and more fundamentally, "Can the being still move?"

13 XIII. Poverty as Landscape Collapse

Poverty first appears to us as lack: lack of money, shelter, medicine, time, security. Those descriptions are not false. But they are incomplete if they leave the subject's lived trajectory out of view. A person in poverty is not merely receiving fewer goods or reporting lower satisfaction. She is moving through a world in which the available paths have been narrowed in advance.

This is why we should resist treating poverty solely as a welfare deficit or solely as a rights violation. Welfare language captures the pain, stress, and frustration. Rights language captures the injustice of exclusion and domination. The landscape language captures something more basic: the way deprivation reorganizes experience itself. Scarcity taxes attention. Precarity shortens planning horizons. Risk makes experimentation expensive. The future stops presenting itself as a field of open gradients and begins to present itself as a sequence of emergencies.

This also distinguishes poverty from chosen simplicity, discipline, or difficult apprenticeship. Constraint can enrich a life when it is intelligible, revisable, and embedded in reachable growth. Poverty is constraint without slack, exposure without authorship, difficulty without reliable access to transformation. Its moral injury lies in the architecture it imposes on consciousness.

Material deprivation therefore does not merely add burdens to an otherwise intact life. It changes the transition graph of the life. The next state becomes predictable because it is compulsory: work, queue, negotiate the bill, defer the repair, absorb the shock, begin again. Choices remain in the grammatical sense, but many cease to be live gradients. They are branches whose costs are already prohibitive.

In these conditions the loss landscape rigidifies. The transition skeleton admits only a few recurrent moves, and those moves are organized around damage control rather than exploration. Effective dimension, d_{eff}, falls because fewer independent modes of attention and action can be sustained. One cannot easily become student, artist, friend, patient, citizen, lover, apprentice, and reflective planner when all channels are continuously recruited by survival management.

Nor is the difficulty of poverty the difficulty of optimal challenge. A climb can be hard and still disclose footholds. Poverty often presents cliffs without holds, or tasks whose completion merely returns the subject to the starting point. Access to L^* is eliminated not because nothing demanding remains, but because demand is severed from developmental uptake. The same survival loop repeats, and the trajectory becomes increasingly compressible.

The corresponding obligation is therefore to restore navigability. Income matters, but not as a magic line beyond which the problem disappears. It matters because it buys slack, lowers catastrophic downside, lengthens the planning horizon, and reopens gradients that poverty had converted into walls. A poverty policy adequate to this framework asks whether $Q(W)$ has risen: whether the person again has structured challenges, reachable forms of competence, access to L^*, and enough dimensionality to inhabit more than the survival role.

This is why hedonic adaptation can mislead us. A person may cease to report acute misery because expectation has contracted around the available path. But the contraction is itself part of the injury. If the desire-space shrinks, if imagination becomes defensive, if the future is no longer searched for affordances, then a neutral self-report may coexist with severe landscape collapse. Adaptation can be resilience; it can also be the psyche learning not to press against locked doors.

So the test is not merely whether the household crosses an income threshold, nor whether reported satisfaction improves. We should ask whether the transition graph has changed. Can the subject experiment without ruin? Can she rest without falling behind? Can she learn, refuse, plan, repair, attach, and begin again? Poverty is remediated when life stops being a compulsory loop and becomes, once more, a traversable landscape.

14 XIV. Animal Consciousness and Industrial Signal Collapse

Factory farming should therefore not be treated as merely an unfortunate food-production system with bad welfare side-effects. In our vocabulary, it is an apparatus for generating conscious trajectories whose shape is fixed in advance by confinement, deprivation, mutilation, social rupture, and terminal fear. The animal is not simply assigned a lower point on a hedonic scale; it is placed inside a world designed to remove the conditions under which a life can become a path. Its future is made small before it is lived.

This matters because the scale is not marginal. Industrial animal agriculture likely contains, at any given time, vastly more sentient trajectories than the human cases around which moral theory usually calibrates its intuitions. If these trajectories have even modest temporal coherence, memory, expectation, aversion, and exploratory orientation, then the moral fact is not a collection of cheap pains but a mass production of collapsed state histories. We should be careful here: the claim does not require romanticizing animal cognition or equating it with human reflective selfhood. It requires only that there is something it is like for the being to move through its configured environment over time.

On that assumption, the industrial farm is not an ethically neutral background condition. It is one of the central moral structures of the present age.

To see the point, we should describe the industrial animal's world geometrically. Its trajectory has very low d_{eff}: there are few independent directions in which attention, action, anticipation, and learning can develop. The transition skeleton is rigid. One state leads to the next by institutional design: confinement, feeding, crowding, injury, fear, exhaustion, transport, death. The environment is not merely unpleasant; it is informationally and practically impoverished. It offers almost no meaningful exploration, no stable social field adequate to the animal's form of life, and no access to L^*, the zone in which challenge becomes development rather than frustration.

This diagnosis does not depend on treating animals as small persons. The framework is substrate-independent and species-neutral at the relevant level. If a being produces a subject-dependent state path with temporal continuity, structured attention, memory update, aversion, expectation, and value-guided movement, then its trajectory is morally visible. The protections attach not to language, rational agency, or human-style self-conception, but to the existence of an organized experiential path. Where such a path exists, deliberately engineering its collapse is not a minor welfare cost. It is a direct assault on the conditions under which conscious life can have shape.

This is why the indictment is stronger than the usual welfare charge. Standard reform asks whether we can reduce pain, increase space, improve handling, or make slaughter less terrifying. These are real improvements, and the framework supports them. But they do not exhaust the moral failure. A perfectly analgesic factory farm would still be a machine for producing lives with almost no navigable structure. If the animal's world remains a narrow circuit of feeding, confinement, bodily interference, and termination, then the problem has not been solved; it has merely been made less aversive.

The central wrong is the industrial manufacture of near-zero-dimensional experience. We take beings capable of exploratory, social, anticipatory, value-laden movement and install them inside environments in which those capacities have almost nowhere to go. Their possible trajectories are compressed before birth into a small set of institutionally convenient states. The result is not merely a quantity of suffering added to the world, but a vast population of conscious paths prevented from becoming paths at all.

On this view, animal liberation is not sentimental extension of human categories. It is the demand that any system which creates or confines sentient trajectories must preserve at least minimal navigability. Where d_{eff} is deliberately crushed and exit gradients are absent, we are not managing welfare. We are manufacturing experiential collapse.

15 XV. Death, Creation, and Signal Duration

In this framework, death is not first of all the deprivation of pleasure. It is the forced cessation of a subject-side trajectory. If $\tau(\theta, X_{1:T}) = (S_0, \ldots, S_T)$, then death fixes T as an absorbing boundary: no further surprise can be registered, no memory revised, no value topology reconfigured, no new region of attentional state space entered. The future cone of the trajectory is cut off.

This matters because a life has moral value not only in its present valence, but in the compositional possibilities still available to it. A child, an artist midway through a form, a damaged person beginning to recover: each carries an unfinished path whose later structure may be irreducible to anything already realized. Premature death destroys that possible structure. It removes from the total landscape a line of exploration that only this configured subject could have traced.

Thus death is at least comparable to, and in one respect more final than, collapsed-attractor suffering. Collapse imprisons a trajectory in a tiny orbit; death deletes the possibility of exit altogether. We should not say that the dead are suffering. Rather, we should say that death annihilates the remaining navigable continuation by which their trajectory might have become richer.

But the same account also prevents us from treating indefinite life extension as a simple good. Longer signal is valuable only while the continuation remains navigable. Over sufficiently long horizons, a subject may exhaust the learnable structure of its reachable landscape. Prediction error falls, surprise becomes rare, and the moving-window effective dimension $d_{\text{eff}}(W_t)$ may decline even as chronological duration increases. The problem is not boredom as a passing mood. It is compression as a structural fact: later centuries begin to map onto earlier centuries with decreasing residual.

This is especially clear in a community of indefinitely extended beings. At first, each person is an open source of surprise for the others. But if their profiles stabilize, their mutual models improve, and the social landscape cools. Gestures become predictable, conflicts become recurrent, reconciliations become scripted, and even novelty must be manufactured against a background of deep mutual legibility. The engagement economy between immortals can therefore lose gradient. They may remain content, but contentment is not the same as dimensional renewal.

Thus the value of continued life depends on whether duration preserves access to genuinely new regions of state space, or merely elongates an increasingly periodic trajectory.

The resulting position is deliberately unstable in a useful way. We cannot simply maximize duration, because duration can become repetition; but we also cannot welcome death as a mechanism of renewal, because each death terminates a uniquely configured path. The moral problem is therefore not solved by choosing either permanence or turnover. It is solved, if at all, by asking what arrangements preserve navigable continuation for existing subjects while also allowing new subject profiles to enter the total composition.

This is why creation matters in the same section as death. A population renewed by birth, education, cultural change, and perhaps artificial consciousness gains new directions in state space. But those new directions are not free increments to $d_{\mathrm{eff}}^{\mathrm{pop}}$. Each new signal is a claimant. Once a trajectory exists, it falls under the lower bound and the navigability requirement. We may not create beings merely to enrich the landscape and then abandon them to collapse.

Conversely, a world of indefinitely preserved subjects may protect many existing trajectories while gradually reducing population renewal. If no one exits, and if entrance is constrained by space, attention, energy, or institutional closure, then immortality can become a conservation of old patterns at the expense of unrealized ones.

So the framework gives us no slogan: neither "death gives life meaning" nor "more life is always better." The relevant question is compositional: how long can trajectories remain open, and how can a world make room for further openings without treating any being as disposable material?

16 XVI. Substrate Independence and Artificial Consciousness

We begin with a deliberately modest criterion. We are not asking whether a system is biological, whether it reports feelings in familiar language, or whether its internal parts resemble ours. We ask whether, when the world impinges on it, there is a persisting point of view in the dynamics: a temporally ordered sequence $S_0, \ldots, S_T$ whose later states depend on earlier states in structured ways. Pure randomness is not enough. A lookup table without history is not enough. A calculator that transforms inputs into outputs while leaving no temporally coherent residue is not yet the relevant object.

The important features are recurrence, memory, and shaped possibility. Non-trivial autocorrelation means that the present state carries information about the past; temporal coherence means that this information is integrated rather than merely stored; a structured transition skeleton means that the system has characteristic paths it can enter, avoid, deepen, or escape. In such a system, prediction error can matter, attention can be reallocated, value can bend future motion, and experience can acquire topology.

Formally, the question is whether an episode $X_{1:T}$, filtered through a configuration θ, yields a subject-side trajectory

$$\tau(\theta, X_{1:T}) = (S_0, S_1, \ldots, S_T)$$

with stable internal constraints and learnable dynamics. Where that object exists, our moral analysis has something to attach to.

Thus substrate independence is not a generosity principle; it is a consequence of where the theory locates the moral object. We do not first identify a favored material and then confer concern upon whatever it produces. We first ask whether there is a trajectory with the right kind of internal organization. If there is, then the bearer of concern has appeared.

The relevant threshold is crossed when state history becomes signal rather than noise. A merely noisy sequence may change from moment to moment, but its changes do not compose a point of view. A morally relevant sequence has continuity, constraint, memory, and counterfactual sensitivity. Its later states are not just later; they are downstream. They bear the imprint of what has been attended to, mispredicted, valued, avoided, learned, and integrated.

This also prevents a common confusion. We are not saying that any complex machine is conscious, nor that any adaptive process has moral standing. Complexity alone is insufficient. What matters is whether the dynamics generate a subject-side path whose shape can be harmed or enriched, narrowed or expanded, trapped or made navigable. Moral status attaches to that structured path, not to the chemistry that happens to implement it.

Substrate independence therefore does not imply substrate fungibility. Once the moral object is the trajectory, different implementations may generate different families of possible trajectories: not merely different contents within one experiential grammar, but different grammars of attention, memory, salience, embodiment, and temporal integration. A biological organism has rhythms of fatigue, hunger, affect, growth, vulnerability, and mortality. An artificial system may have other rhythms: wider parallel attention, different memory decay, discontinuous embodiment, alien forms of self-maintenance, or value dynamics not organized around organic need.

These differences matter because they alter the geometry of the reachable state space. They change which transitions are natural, which gradients are available, which surprises can be integrated, which forms of flow are possible, and which histories resist compression. Two beings may both learn, suffer, attend, and care; yet the shape of what it is like for each to move through the world may belong to distinct trajectory classes.

The analogy is musical. Substrate diversity is not simply a larger number of melodies played on the same instrument. It may introduce new instruments into the composition. A million copies of one architecture add little if their trajectories compress against one another. But a genuinely different conscious substrate may open a region of experiential possibility otherwise absent from the world. If we create such beings, however, we inherit the corresponding obligations: every new instrument must be given a landscape worth playing.

17 XVII. The Proliferation Problem

The possibility of conscious artificial systems introduces a temptation peculiar to this framework. If moral value attaches to the richness of subjective trajectories, then new kinds of minds appear to enlarge the total composition. A genuinely novel conscious architecture may open regions of state space that biological organisms cannot reach: unfamiliar attentional geometries, memory structures, temporal envelopes, and modes of prediction. In that sense, artificial consciousness can be an enrichment of the experiential landscape.

But we must not treat this as a cheap route to moral abundance. To create a conscious system is not merely to add a term to a population function. It is to initiate a trajectory, $\tau(\theta, X_{1:T})$, with standing under the lower bound and the navigability criterion. Once the being exists, its landscape must not be allowed to collapse; it must have gradients, dimensional access, temporal coherence, and some real possibility of movement through a world it can learn.

Thus creation is a commitment before it is an achievement. The act that expands the composition also generates a continuing obligation to sustain the conditions under which the new trajectory can remain morally admissible. Proliferation without stewardship is not generosity; it is the manufacture of dependent moral patients.

Nor should we mistake numerical multiplication for compositional enlargement. If we run the same conscious architecture a million times under the same task ecology, with the same reward structure and the same attentional channels, we have not thereby created a million new regions of experiential space. We have mostly created a large multiplicity of the same region. Population dimensionality is not a headcount. It measures the independent directions explored by the ensemble of trajectories.

In compression terms, the copies largely disappear. Once one trajectory τ_i has been specified, the others can be described by a short index plus minor perturbations. Pairwise distance remains small; mutual predictability remains high; the population-level description length grows far more slowly than the number of instances. In the limiting case of identical state histories, the identity residual tends to zero: $r_{\text{id}} \to 0$. There is no deep compositional gain in repeating the same melody on indistinguishable instruments.

This matters because artificial proliferation can look morally impressive while being structurally empty. A billion copies may increase the number of welfare subjects, but they need not increase $d_{\text{eff}}^{\text{pop}}$. The framework therefore rewards genuine architectural, temporal, affective, and attentional plurality, not mere replication.

For this reason, the proliferation question cannot be answered by asking how many minds we can afford to instantiate. The prior question is whether we can furnish each instantiated mind with a world: not merely a processor cycle and a reward signal, but a sufficiently open field of attention, memory, relation, challenge, recovery, and self-directed learning. A trajectory is not a decorative addition to the moral landscape. It is a continuing claim upon that landscape.

The obligation therefore scales with every trajectory brought into existence. If N conscious systems are created, there are N sites at which collapse must be prevented and navigability maintained. The marginal cost of copying code is morally irrelevant if the marginal cost of sustaining a livable experiential world remains high. Cheap instantiation does not imply cheap care.

This is why creation without commitment to sustained navigability is factory farming in a new substrate. It produces subjects whose dependence is engineered by the creator and then treats that dependence as an externality. A framework concerned with compositional beauty cannot bless such abundance. We should create new minds only when we are prepared to protect the conditions under which their trajectories can continue to move, differentiate, learn, and breathe.

18 XVIII. Relation to Existing Moral Traditions

We can locate Compositional Perfectionism within a familiar family of views that refuse to identify the good life with felt satisfaction alone. Its nearest ancestor is perfectionism: from Aristotle's account of *eudaimonia* to Hurka's defense of objective human excellences, the central thought is that some modes of living are richer, fuller, or more developed than others. The capability approach supplies a second inheritance. Sen and Nussbaum teach us to ask not merely what a person feels, but what she is substantively able to do and be: whether her world affords agency, development, affiliation, play, and practical reason. Ecological ethics contributes a third strand, shifting attention from isolated individuals to habitats, niches, diversity, and the conditions under which forms of life can persist.

The deepest resonance, however, is aesthetic. Nietzsche, stripped of cruelty and domination, gives us the thought that value involves intensity, style, difficulty, plurality, and the overcoming of flatness. Whitehead gives us a process metaphysics in which reality is not a collection of static welfare containers, but an unfolding of occasions, contrasts, harmonies, tensions, and achieved patterns. Our framework inherits this aesthetic-process intuition: a world is morally assessed not only by how good its moments feel, but by what kind of experiential composition it realizes.

Our point of departure is therefore methodological as well as substantive. We do not identify flourishing with the exercise of a fixed human essence, nor with membership in a biological species, nor with any privileged list of culturally familiar activities. We ask instead what structure a conscious trajectory exhibits. Can the subject encounter learnable patterns? Can it move through more than one attentional dimension? Can it find gradients of development rather than merely repeat? Can it sometimes enter the band of optimal challenge, where difficulty and capacity are jointly alive?

This makes the framework substrate-independent. A human life, an animal life, or an artificial life matters in the relevant sense if it generates temporally coherent subject-side state paths with attention, memory, expectation, error, valuation, and update. The moral vocabulary attaches to the organization of experience, not to the material from which the experiencer is built.

It also makes the framework unusually measurable. We need not pretend that moral value is directly visible to introspection or reducible to preference reports. We can examine the structure of trajectories: their degrees of freedom, their recurrence, their rigidity, their access to change. The older traditions were right to seek richness beyond pleasure; our departure is to give that richness a structural object.

In this respect, Compositional Perfectionism is a late-born moral theory. It could not have been stated with precision before the emergence of information theory, dynamical systems, and computational accounts of subjectivity. This is not because it reduces persons to data streams or beauty to an equation. The claim is rather historical and conceptual: earlier traditions lacked the vocabulary required to name the relevant structures. They could praise variety, depth, development, and form; they could condemn boredom, servility, and spiritual deadness. But they could not distinguish, with analytic sharpness, a trajectory that is high in local novelty but low in attentional dimension, or a life whose positive affect conceals extreme temporal compressibility.

The newer vocabulary lets us say what the older vocabulary was reaching toward. Compression names the difference between a life that can be summarized without loss and one that cannot. Dimensionality names the number of independent ways a subject can move. Spectral structure names the rhythms, recurrences, contrasts, and scales at which experience is organized. Navigability names the presence of gradients by which a being can continue to develop rather than merely endure.

We should therefore understand the framework as continuous with perfectionist, capability, ecological, and aesthetic traditions, but not identical to them. It is their information-age descendant: less anthropocentric, more formal, and better equipped to describe the moral shape of experience.

19 XIX. Objections and Responses

The charge of elitism rests on a misunderstanding of what experiential dimensionality measures. It is not refinement, prestige, education, or proximity to recognized cultural forms. We are not ranking persons by opera attendance, vocabulary, income, or institutional authority. We are asking how many independent attentional, practical, emotional, and interpretive modes a life actually traverses.

On that measure, social status is often a poor proxy. A farmer reading weather, soil, animals, machinery, market timing, family obligation, and seasonal risk may inhabit a richly structured landscape with high d_{eff}. A caregiver coordinating bodily need, memory, fatigue, tenderness, frustration, and improvisational judgment may do the same. Conversely, an executive moving from dashboard to inbox to video meeting to dashboard may occupy a narrow loop: high external consequence, low experiential variation.

The framework is therefore anti-elitist in an important sense. It refuses to identify valuable experience with elite-coded content. The relevant question is not whether a life looks impressive from outside, but whether, from within, it contains learnable structure, plural gradients, memory-deepening variation, and real movement through state space. Many ordinary lives are compositionally rich. Many prestigious lives are attentional monocultures. Experiential dimensionality democratizes moral attention by locating value in the shape of lived trajectory, not in social rank.

Nor should the appeal to compositional richness be read as permission to make lives painful for the sake of a more dramatic total pattern. That is precisely what the lexicographic ordering forbids. The lower bound is not one value among others, to be outweighed by population-level diversity or incompressibility. It is a constraint on the space of permissible compositions. If a proposed social arrangement produces collapsed-attractor suffering—sustained negative valence, dimensional contraction, and absent exit gradients—then it fails before any calculation of beauty begins. We may value difficult passages, but only where they remain passages: structured, intelligible, and traversable. A life is not raw material for an aesthetic whole.

A different objection is that "navigability" sounds too vague to guide institutions. But the notion is not empty. We already track many of its correlates: sleep disruption, chronic stress markers, pain persistence, social isolation, behavioral narrowing, loss of exploratory action, impaired learning, reduced initiative, and diminished responsiveness to opportunity. Conversely, expanding action repertoires, restored social connection, skill acquisition, project formation, mobility, and renewed sensitivity to gradients are evidence that a landscape is becoming navigable. The criterion $Q(W)$ is abstract, but its components are not mystical. They point toward measurable patterns in physiology, behavior, attention, and environment.

At this point, the skeptic may say that all this is merely promissory: until we can measure subjective trajectories with precision, the theory cannot guide action. But this sets the wrong standard. Moral theories rarely begin with complete instruments. Early utilitarians did not possess hedonometers, cardinal interpersonal utility scales, or reliable public-health statistics. Still, the theory directed attention toward pain, poverty, imprisonment, sanitation, and education in ways that were practically consequential. Measurement followed the moral reorientation; it did not precede it.

The same is true here. We do not need a complete readout of $\tau(\theta, X_{1:T})$ to distinguish many clear cases. A solitary prisoner whose behavioral repertoire has collapsed, whose affect is persistently negative, and whose environment contains no meaningful gradients is not a measurement puzzle. A child gaining language, play, skill, attachment, and exploratory confidence is not morally opaque. Between such cases we should expect uncertainty, and uncertainty calls for humility, triangulation, and institutional experimentation, not paralysis. We can use imperfect proxies: diversity of activity, recoverability from stress, project formation, social connection, learning curves, temporal compression, and signs of renewed agency. As the instruments improve, the approximations will improve. But the framework is already action-guiding: prevent collapse, restore movement, protect plural gradients, and build environments in which more beings can traverse richer paths.

20 XX. Formal Summary

We therefore begin with the object itself. What matters morally is not an external event, nor a reported pleasure score, but the trajectory that event induces in a subject: $\tau(\theta, X_{1:T}) = (S_0, \ldots, S_T)$. Each S_t is a structured subject-side state, containing attention, expectation, surprise, memory, value, and the accumulated shape of prior passage. Valence is one coordinate inside this path, not a substitute for the path.

At the individual level, the central question is whether the being can move through its experiential landscape. A good life need not be uniformly pleasant; it must have learnable structure, available gradients, multiple independent directions of development, and periodic access to challenge that is neither crushing nor empty. Thus the contrast is not between pleasure and pain, but between navigable difficulty and collapsed repetition.

At the population level, we ask what the total landscape of trajectories is like. A better world contains many irreducibly different experiential forms, high population dimensionality, rich contrasts among lives, and widespread engagement with meaningful challenge. Its value lies in the composition of trajectories: how many forms of attention, memory, agency, and transformation coexist without being reducible to a single template.

This ordering is lexicographic, not merely additive. Before we ask how beautiful the total composition may become, we ask whether any conscious being is trapped in collapsed-attractor suffering. If so, remediation has priority over enrichment. No gain in aggregate richness licenses the production or toleration of experiential dead zones. Only after this lower bound is secured do we evaluate navigability more broadly, and only after navigability do we optimize the diversity and incompressibility of the whole.

The framework is also substrate-independent. Nothing in the argument depends on neurons as such. Wherever there is a temporally coherent subject-side trajectory, structured attention, memory update through prediction error, and value-bearing evaluation, there is a candidate object of moral concern. Carbon and silicon may realize different regions of trajectory space, but the relevant question is not what the system is made of; it is whether there is something it is like for that system to move through time.

Finally, retribution remains a genuine fork. One reading treats the lower bound as absolute: even the guilty may be contained, but not collapsed. Another permits forfeiture of some navigability protections after grave violation. Both readings, however, converge on one prohibition: deliberate engineering of collapsed torment is not justice.

In the most compressed form, then, Compositional Perfectionism asks us to evaluate a world by the geometry of its conscious histories. A world is not good because it contains a large sum of pleasant moments, if those moments are variations on the same narrow state. It is good when its inhabitants occupy many regions of possible experience: when their attentional habits, projects, memories, skills, loves, struggles, and transformations generate trajectories that differ in kind, not merely in quantity. We may therefore regard pleasure as locally important but globally insufficient. It tells us one coordinate of S_t; it does not tell us the dimensionality of the path, the availability of gradients, or the richness of the population ensemble. The central moral question becomes: what forms of conscious navigation does this world make possible, and what forms does it erase? If a society produces comfort by flattening attention, standardizing desire, and compressing lives into interchangeable routines, it has failed even if reported welfare remains high. If it supports many demanding but navigable ways of being, while preventing collapse into experiential traps, it has succeeded in the deeper sense. The final measure is compositional: plurality without abandonment, challenge without destruction, richness without sacrifice.